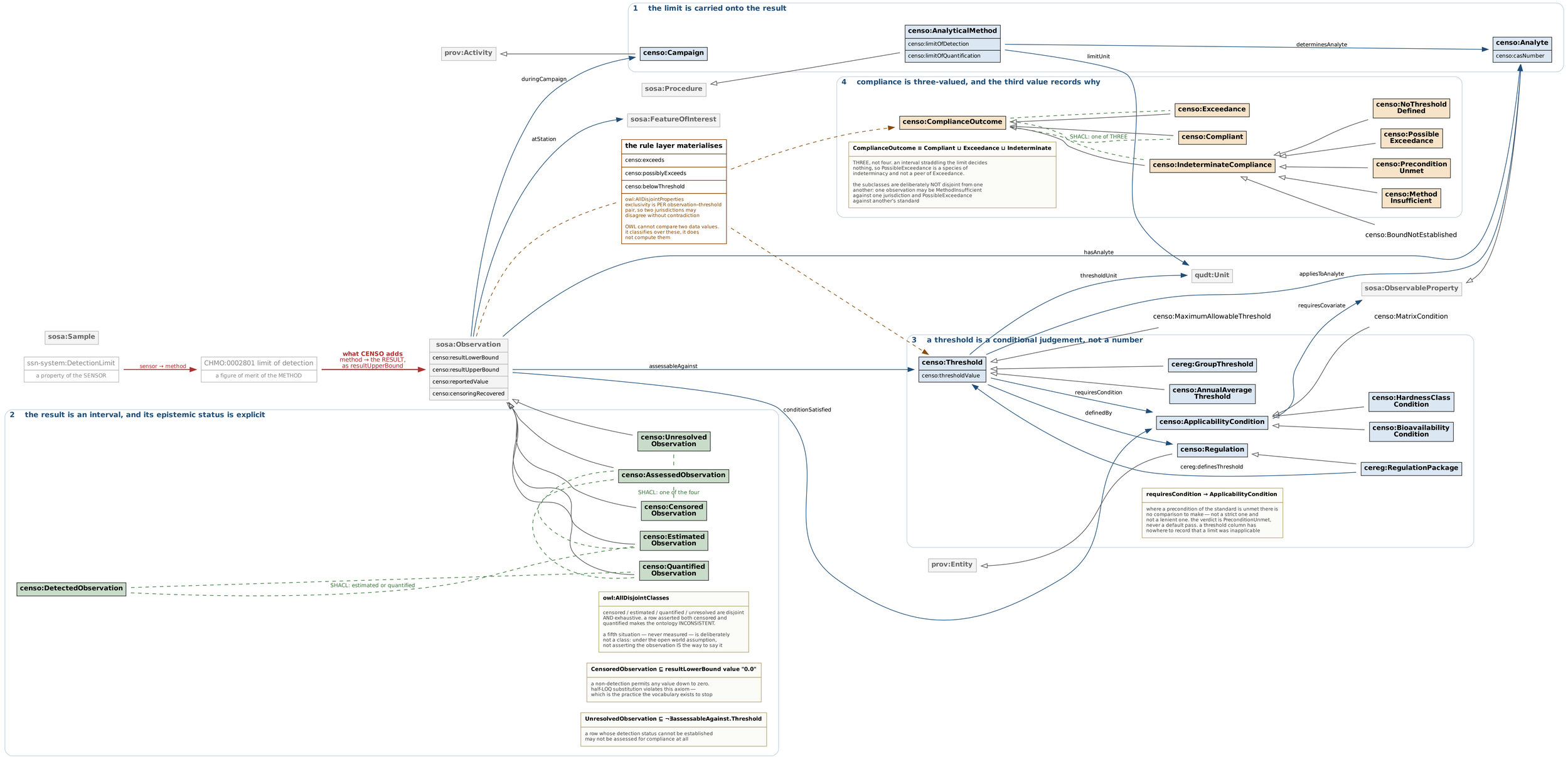

1 the limit is carried onto the result
censo:AnalyticalMethod
censo:limitOfDetection
censo:limitOfQuantification
determinesAnalyte
censo:Analyte
censo:casNumber
limitUnit
prov:Activity
censo:Campaign
duringCampaign
sosa:Procedure
sosa:FeatureOfInterest
atStation
4 compliance is three-valued, and the third value records why
censo:ComplianceOutcome
censo:Exceedance
censo:Compliant
censo:IndeterminateCompliance
SHACL: one of THREE
censo:NoThreshold Defined
censo:Possible Exceedance
censo:Precondition Unmet
censo:Method Insufficient
censo:BoundNotEstablished
ComplianceOutcome ≡ Compliant ⊔ Exceedance ⊔ Indeterminate
THREE, not four. an interval straddling the limit decides nothing, so PossibleExceedance is a species of indeterminacy and not a peer of Exceedance.
the subclasses are deliberately NOT disjoint from one another: one observation may be MethodInsufficient against one jurisdiction and PossibleExceedance against another's standard
the rule layer materialises
censo:exceeds
censo:possiblyExceeds
censo:belowThreshold
owl:AllDisjointProperties
exclusivity is PER observation–threshold pair, so two jurisdictions may disagree without contradiction
OWL cannot compare two data values. it classifies over these, it does not compute them
hasAnalyte
thresholdUnit
qudt:Unit
appliesToAnalyte
sosa:ObservableProperty
requiresCovariate
censo:MaximumAllowableThreshold
censo:MatrixCondition
sosa:Sample
ssn-system:DetectionLimit
a property of the SENSOR
sensor → method
CHMO:0002801 limit of detection
a figure of merit of the METHOD
what CENSO adds
method → the RESULT, as resultUpperBound
sosa:Observation
censo:resultLowerBound
censo:resultUpperBound
censo:reportedValue
censo:censoringRecovered
assessableAgainst
3 a threshold is a conditional judgement, not a number
censo:Threshold
censo:thresholdValue
cereg:GroupThreshold
censo:AnnualAverage Threshold
requiresCondition
definedBy
censo:ApplicabilityCondition
censo:HardnessClass Condition
censo:Bioavailability Condition
censo:Regulation
cereg:definesThreshold
cereg:RegulationPackage
requiresCondition → ApplicabilityCondition
where a precondition of the standard is unmet there is no comparison to make — not a strict one and not a lenient one. the verdict is PreconditionUnmet, never a default pass. a threshold column has nowhere to record that a limit was inapplicable
prov:Entity
conditionSatisfied
2 the result is an interval, and its epistemic status is explicit
censo:Unresolved Observation
censo:AssessedObservation
SHACL: one of the four
censo:Censored Observation
censo:Estimated Observation
censo:Quantified Observation
censo:DetectedObservation
SHACL: estimated or quantified
owl:AllDisjointClasses
censored / estimated / quantified / unresolved are disjoint AND exhaustive. a row asserted both censored and quantified makes the ontology INCONSISTENT.
a fifth situation — never measured — is deliberately not a class: under the open world assumption, not asserting the observation IS the way to say it
CensoredObservation ⊑ resultLowerBound value "0.0"
a non-detection permits any value down to zero. half-LOQ substitution violates this axiom — which is the practice the vocabulary exists to stop
UnresolvedObservation ⊑ ¬∃assessableAgainst.Threshold
a row whose detection status cannot be established may not be assessed for compliance at all